

Edge client k

Sensor
stream

Local training
(full model federated)

Prediction
residuals

send

EF top- k
+ 8-bit quant

Event trigger
 $\|\Delta_k\| < \gamma H_k$?

Page-Hinkley
drift detector

upload $(\Delta_k, n_k, d_k, a_k)$

broadcast global model θ

Server

Decompress; staleness- &
drift-aware weights + adaptive
clip $c \cdot \text{med } \|\Delta\|$

Aggregate $\rightarrow$ new
global model θ

d_k